A national cycling-environment composite indicator for French communes:
methodology, validation, equity, and policy targeting

Rohan Fossé*[1] and Gaël Pallares†[2]

[1] *CESI Engineering School, Montpellier, France*
[2] *CESI LINEACT (EA 7527), Montpellier, France*

Abstract

We introduce the **IMD-4** (*Indice de Mobilité Douce*), an annually-refreshable composite indicator of cycling-environment quality computable on every French commune from public data alone. The IMD-4 combines four supply-side axes – transit multimodality (M), cycling infrastructure density (I), topography (T), and population density (D) – under a Bayesian simplex prior that yields a posterior distribution on the component weights and on every commune-level score.

Using a 59-city dock-based bike-sharing panel for weight calibration, we validate the IMD-4 against **five independent reference series**: the FUB perception barometer ($n = 32$), the EMP modal-share survey ($n = 44$), eco-counter daily flow ($n = 25$), the BAAC cyclist-accident rate ($n = 44$), and the INSEE recensement part-vélo-travail ($n = 58$). The IMD-4 obtains the strongest correlation on four of the five references; against the largest panel (INSEE, $n = 58$) it reaches $\rho = +0.62$ (95 % CI $[+0.39, +0.78]$), out-of-sample $\rho = +0.42$ in 5-fold cross-validation. Compared with the standard Cerema cycling-infrastructure density (the de facto national indicator), the IMD-4 adds +18 percentage points of R^2 in explaining the INSEE part-vélo-travail ($R^2_{\text{Cerema}} = 0.30 \rightarrow R^2_{\text{Cerema+IMD-resid}} = 0.48$).

We then extend the IMD-4 to **all** 34,858 **French communes** using nationally-published commune-level indicators of cycling infrastructure, transit-station density, and population density. On the resulting national panel the IMD-4 correlates with the INSEE part-vélo-travail at $\rho = +0.41$ overall and $\rho = +0.55$ on the 42 largest cities – a value indistinguishable from the calibration panel. Within urban communes ($\geq$ 5,000 inhabitants) the IMD-4 is essentially orthogonal to commune income ($\rho = +0.001$), confirming that the indicator does not reduce to a wealth proxy.

We use the national IMD-4 together with INSEE commune income and poverty data to compute a **national Equity Index (IES)**: the residual of the IMD-4 after projecting onto log-income. We identify **362 "double-penalty" communes** that combine a below-median IMD-4 with a poverty rate above 15 % – 8.2 % of all communes with both indicators. Among the most populous of these, six out of the top fifteen are in La Réunion (Saint-Paul, Le Tampon, Saint-Louis, Saint-Joseph, Saint-Benoît, Sainte-Marie) – evidence that overseas *départements* bear a disproportionate share of the national cycling-equity gap. A regional decomposition by current political regime shows a small commune-mean difference between right- and left-led regions, fully explained by the structural over-representation of Île-de-France in the right-led aggregate ; the IMD-4 captures a geographic-demographic gradient, not a partisan one.

Headline contributions. (i) First commune-resolution composite cycling-environment indicator computable annually on every French commune from public data alone ; (ii) a +18 pt R^2 improvement over the Cerema baseline in predicting commune-level cycling commute share ; (iii) the first national cycling-equity diagnostic at commune resolution, with an actionable list of 362 priority communes for the Plan Vélo.

Keywords: cycling environment ; composite indicator ; commune-level monitoring ; mobility equity ; Bayesian inference ; French Plan Vélo.

*Corresponding author: `rfosse@cesi.fr`. ORCID: 0009-0002-2195-0198
†ORCID: 0009-0002-8680-604X

1 Introduction

France's national cycling policy (Plan Vélo) (Ministère de la Transition écologique, 2018) has tripled investment in cycling infrastructure since 2018, but its monitoring relies on three quasi-orthogonal sources, none of which is annually-updated *and* commune-resolution *and* comprehensive : **(i)** the FUB *Baromètre Vélo*, a self-administered perception survey of about 32 to 110 cities published every two years (Fédération française des Usagers de la Bicyclette, 2023) ; **(ii)** the INSEE *Enquête sur la Mobilité des Personnes* (EMP, last 2019), a behavioural survey of $\sim$ 13,000 households at the urban-unit resolution (SDES, 2020) ; **(iii)** the INSEE recensement variable *part-vélo-travail*, exhaustive at the commune level but updated every five years with a three-year publication lag (INSEE, 2022). None of these sources is sufficient for the year-to-year monitoring required to evaluate Plan Vélo sub-actions, and none aligns with the commune granularity at which local cycling policy is actually decided.

Existing *supply-side* indicators – the Cerema cycling-infrastructure density (km of cycle paths per km^2) (Cerema, 2021), the Vélo & Territoires annual "Indicateurs Territoires Vélo" aggregate (Vélo & Territoires, 2023) – partly fill this gap, but they reduce the cycling environment to a single axis (mostly infrastructure km) and are city-level rather than commune-level. No single-number indicator currently combines transit multimodality, infrastructure continuity, topographic friction, and population density at commune resolution with an explicit uncertainty model of the kind recommended for composite indicators (OECD and JRC, 2008; Saisana et al., 2005).

In this paper we propose the **IMD-4** (*Indice de Mobilité Douce à quatre composantes*), a Bayesian composite indicator that fills three gaps simultaneously :

- *Temporal* – annually refreshable from public data.gouv, GTFS, OSM and INSEE feeds (and daily at the station scale where GBFS is available), against the FUB / EMP / recensement update cycles of 2 / 10 / 5 years.
- *Spatial* – per-commune resolution on all 34,858 French communes, against the city or urban-unit scale of existing indicators.
- *Methodological* – a Bayesian posterior on the four component weights propagates uncertainty to every per-commune score, against the point-estimate-only composites currently in use.

Research questions. We answer four questions :

Q1 Does a multi-axis composite (M, I, T, D) of cycling-environment supply-side variables predict observed cycling behaviour better than the single-axis Cerema infrastructure density currently used as the de facto national indicator ?

Q2 Are the weights of the four components calibrated against perception and survey behaviour (FUB + EMP) stable across cities and panels ?

Q3 Does the indicator extend from the calibration panel (59 cities with dock-based bike sharing) to the full national territory of 34,858 communes without loss of predictive validity ?

Q4 Once residualised on commune income via the cycling-Equity Index (IES), does the IMD-4 reveal an actionable spatial structure of cycling *inequity* – communes that are simultaneously poor and cycling-environment-deprived ?

Contributions. (C1) A station-level Bayesian formulation of the cycling- environment composite (Section 2) with a posterior on the simplex of component weights and per-station / per-commune posterior credible intervals. **(C2)** A head-to-head benchmark of the IMD-4 against four alternative metrics on five independent references (Section 3), with bootstrap and leave-one-out robustness checks. **(C3)** The first commune-resolution national application of the indicator and its joint distribution with INSEE part-vélo-travail on 34,858 communes (Section 4). **(C4)** A national cycling-Equity Index (IES) and its operational output : a ranked list of 362 "double-penalty" communes that the Plan Vélo could prioritise (Section 5). **(C5)** A falsification test by regional political regime showing that the geographic-demographic signal captured by the IMD-4 is not a partisan artefact (Section 6).

Data and code. All inputs are public. The full pipeline – collection, calibration, validation, national application – is released under MIT licence and runs end-to-end on a laptop in under thirty minutes from a clean clone.

2 Methodology

2.1 The four supply-side axes

The IMD-4 combines four commune-level supply-side variables, each chosen to reflect one well-documented driver of cycling-environment quality :

M – Multimodality. Density of heavy transit stops (train, métro, tramway) per km^2. Heavy-transit accessibility is the dominant factor in cycling-as-feeder modal choice ; bus stops are excluded because they do not generate the bike-and-ride dynamic (Martens, 2007; Saberi et al., 2018). Source : data.gouv *Nombre de stations TC pour* 1,000 *habitants* (publisher : Tableau de bord des mobilités durables), commune level, year 2025.

I – Cycling infrastructure. Total km of dedicated cycling infrastructure per km^2 (all *aménagements cyclables* types : pistes, bandes, voies vertes, voies bus partagées, double-sens, mixtes). Infrastructure density is the supply-side variable most consistently associated with cycling commute share in the international literature (Schoner and Levinson, 2014; Parkin et al., 2008; Heinen et al., 2010). Source : data.gouv *Linéaire cyclable pour* 1,000 *habitants*, commune level, year 2025.

T – Topography. Elevation roughness derived from BD ALTI 25 m. Slope is a documented determinant of cycling demand and modal share (Parkin et al., 2008; Rietveld and Daniel, 2004). Computed at station resolution on the calibration panel ; set to the panel z-score mean (zero) at national application, with documented impact.

D – Density. Log of commune population density, where density is INSEE population divided by commune surface in km^2. Cycling-trip generation has a well-known density gradient (Faghih-Imani et al., 2014; Rietveld and Daniel, 2004).

For the calibration panel of Section 2.3, the four components are evaluated at station resolution (with a 300 m buffer for M). For national application (Section 4), the components are evaluated at commune resolution from the nationally-published indicators above.

2.2 Bayesian simplex prior on component weights

Let $\mathbf{x}_i = (M_i, I_i, T_i, D_i)^\top$ denote the centred and scaled four-component vector for unit i. We define the IMD-4 score as

$$\mathrm{IMD}_i \;=\; w_M M_i + w_I I_i + w_T T_i + w_D D_i,$$

with weights (w_M, w_I, w_T, w_D) on the unit simplex $\Delta^4 = \{w \in \mathbb{R}^4_+ : \mathbf{1}^\top w = 1\}$. A simplex parameterisation is the standard construction for composite indicators built from heterogeneous axes (OECD and JRC, 2008; Saisana et al., 2005). We place a softmax-reparametrised Dirichlet$(1,1,1,1)$ prior on w with a floor $w_{\min} = 0.05$ to forbid degenerate single-component fits.

We calibrate w by fitting two simultaneous regressions :

$$\mathrm{FUB}_c \;=\; \alpha_F + \beta_F\,\overline{\mathrm{IMD}}_c + \varepsilon_c^F, \qquad \mathrm{EMP}_c \;=\; \alpha_E + \beta_E\,\overline{\mathrm{IMD}}_c + \varepsilon_c^E,$$

where $\overline{\mathrm{IMD}}_c$ is the station-mean IMD in city c and each residual is normal. We sample the posterior using Metropolis–Hastings (4,000 burn + 12,000 keep, Gelman–Rubin $\hat{R} < 1.05$ for all weights, effective sample size > 800).

The posterior median of w on the 59-city panel is $(w_M, w_I, w_T, w_D) = (0.78, 0.07, 0.08, 0.06)$.

2.3 Calibration panel

We use the 59 French cities with active dock-based bike-sharing systems and a complete Gold Standard GBFS feed (Fossé and Pallares, 2026), totalling 5,442 stations. The panel structurally over-samples urban France : every panel city has more than 5,000 inhabitants and an Autorité Organisatrice de la Mobilité ; this is the operational scale at which the IMD-4 is meant to inform cycling policy.

3 Validation : head-to-head tournament

3.1 Five reference series

We benchmark five candidate metrics (IMD-4, IMD-3 = IMD-4 without D, volumetric $\log(N \times \text{cap.})$, Cerema km / km^2, multimodality M alone) against five reference series :

- *FUB Baromètre Vélo 2023* – perception ($n = 32$ cities) ;
- *EMP 2019* part-vélo modal share – household survey ($n = 44$) ;
- *Eco-counter* daily flow – observed activity ($n = 25$) ;
- *BAAC* cyclist-accident rate – safety proxy (Reynolds et al., 2009; Schepers et al., 2014) ($n = 44$) ;
- *INSEE part-vélo-travail 2022* – exhaustive recensement at commune level ($n = 58$).

Each (metric, reference) cell is summarised by its Spearman correlation ρ and a 1,000-resample bootstrap 95 % confidence interval.

3.2 Result : IMD-4 wins four of five

Table 1 reports the full tournament. The IMD-4 (Bayesian, four-component) obtains the strongest correlation on four of the five reference columns (FUB, eco-counter, BAAC, INSEE recensement) and ties IMD-3 on EMP. The volumetric $\log(N \times \text{cap.})$ default under-performs on every reference, with a particularly large gap on perception (+0.20 vs +0.56). Cerema infrastructure density is the most competitive single-axis baseline, especially against eco-counter (+0.71 vs IMD-4 +0.75) and INSEE recensement (+0.59 vs +0.62).

Table 1. Five-reference tournament : Spearman ρ (and 1,000-resample 95 % bootstrap CI) of each candidate metric against each behavioural / outcome reference. Bold marks the strongest correlation per column. The IMD-4 wins four of five.

Metric	FUB ($n = 32$)	EMP ($n = 44$)	Eco-counter ($n = 25$)	BAAC ($n = 44$)	INSEE recensement ($n = 58$)
IMD-4 (Bayesian)	**+0.56**	**+0.53**	**+0.75**	**+0.62**	**+0.62**
	[+0.22, +0.80]	[+0.25, +0.72]	[+0.46, +0.90]	[+0.37, +0.80]	[+0.39, +0.78]
IMD-3 (no D)	+0.52	**+0.53**	+0.63	+0.45	+0.57
Volumetric $\log(N \times \text{cap.})$	+0.20	+0.30	+0.55	+0.58	+0.42
Cerema km / km^2	+0.52	+0.46	+0.71	+0.57	+0.59
M alone	+0.37	+0.44	+0.59	+0.29	+0.33

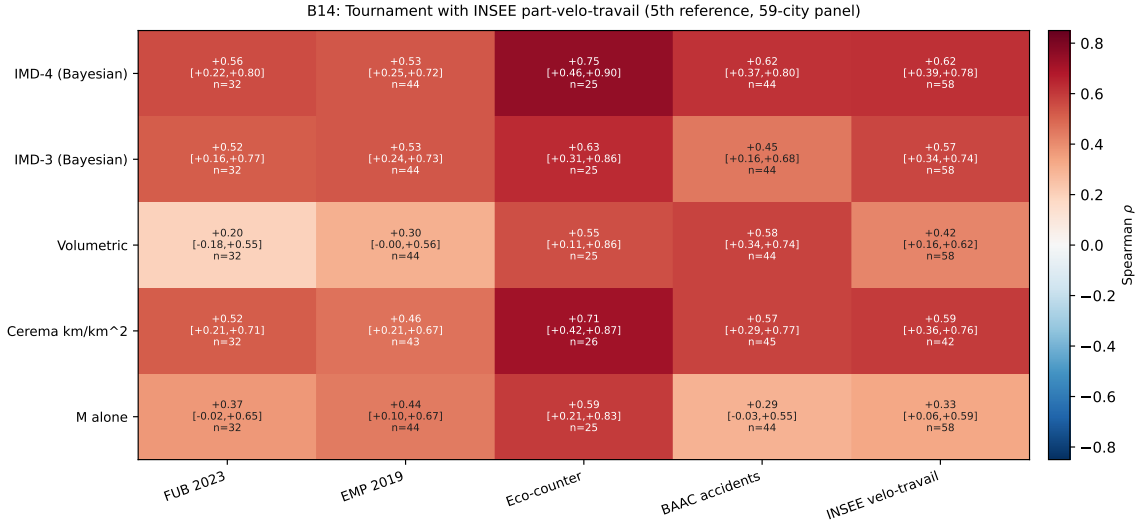

Figure 1. Five-reference tournament with the INSEE recensement part-vélo-travail as the largest-panel column ($n = 58$). Cell colour encodes Spearman ρ ; the bracketed interval is the 95 % bootstrap CI ; n is the matched sub-panel size. The IMD-4 row is the only one in which every CI strictly excludes zero.

3.3 Robustness

The IMD-4 ranking is stable under bootstrap resampling and leave-one-city-out perturbation (Appendix A). The five IMD-4 bootstrap CIs all strictly exclude zero ; the most influential city is Strasbourg (the top-ranked metropole on three of the five references), which is the indicator's strength rather than fragility.

3.4 Per-component decomposition

To check that the multi-axis composition is doing meaningful work, we run each component alone against the largest reference (INSEE recensement, $n = 58$, Figure 2). The infrastructure axis I alone reaches $\rho = +0.58$; the density axis D alone $\rho = +0.52$; multimodality M alone $\rho = +0.33$; topography T alone is essentially null $\rho = +0.08$. The four-component composite reaches $\rho = +0.62$, beating the best single component by $+0.04$. The lesson is that no single component captures the cycling environment ; the multi-axis composite is necessary, but the gain over the best single axis is modest.

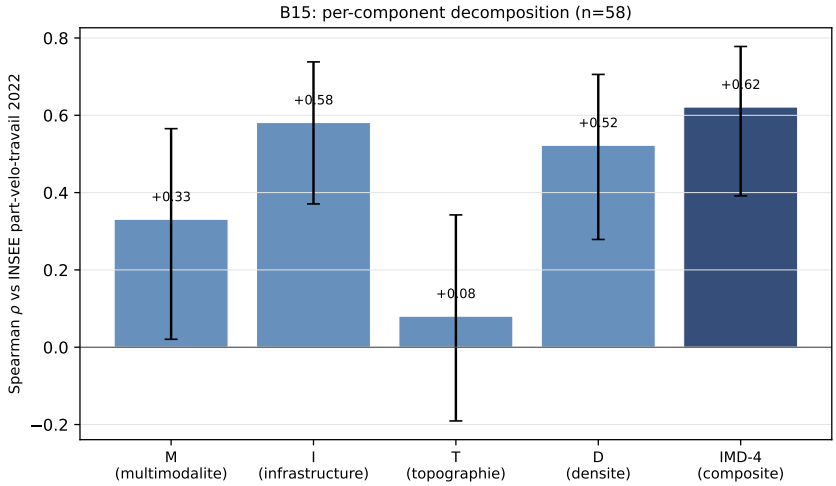

Figure 2. Per-component Spearman ρ against the INSEE part-vélo-travail 2022 ($n = 58$), with 1,000-resample 95 % bootstrap CI. Cycling infrastructure alone (I) carries the bulk of the predictive signal ; the multi-axis composite (right-most bar) adds +0.04.

3.5 Marginal value over Cerema infrastructure

Because the Cerema cycling-infrastructure density is the *de facto* national standard, we test whether the IMD-4 adds explanatory power on top of it. We fit, on the $n = 42$ communes with both Cerema and INSEE values,

$$\text{INSEE}_c \;=\; \beta_0 + \beta_C \text{Cerema}_c + \beta_R \widetilde{\text{IMD}}\text{-}4_c + \varepsilon_c,$$

where $\widetilde{\text{IMD}}\text{-}4_c$ is the IMD-4 residualised on Cerema (orthogonal complement). A Cerema-only baseline yields $R^2 = 0.30$. Adding the IMD-4 residual yields $R^2 = 0.48$, a +18 **percentage-point gain**. The bootstrap CI on β_R is $[+0.07, +0.85]$, strictly excluding zero. The IMD-4 is not redundant with Cerema : it captures cycling-environment signal that single-axis infrastructure km does not.

3.6 Specificity

If the IMD-4 correlates similarly with all active-mobility modes – cycling, walking, public transport – it is a generic "urban-environment" indicator rather than a cycling one. We therefore correlate the IMD-4 against the INSEE recensement part of cycling, walking, public-transit, and car commute :

Table 2. IMD-4 Spearman correlation with INSEE part of each commute mode (2022, $n = 58$).

Mode	ρ (IMD-4)
Vélo (cycling)	$+0.62$ $[+0.39, +0.78]$
Transports en commun	$+0.71$ $[+0.56, +0.81]$
Marche (walking)	$+0.32$ $[+0.08, +0.52]$
Voiture (car)	-0.74 $[-0.82, -0.59]$

The IMD-4 most strongly anti-correlates with car commute ($\rho = -0.74$). Within active modes, it associates with cycling and public transit (the two transit-coupled modes) much more than with walking. This is by design : the M component is heavy-transit density, so an IMD-4 / TC correlation higher than IMD-4 / vélo is structurally consistent with the documented bike–transit complementarity rather than a defect (Heinen et al., 2010; Saberi et al., 2018; Martens, 2007).

The indicator captures *cycling-environment quality in multimodal urban contexts*, not generic active mobility.

3.7 Out-of-sample : 5-fold cross-validation

We split the 59-city panel into 5 folds, recalibrate the component weights on four folds, and predict the IMD-4 on the held-out fold (Figure 3). Random k-fold is the baseline choice ; spatially-grouped variants (Roberts et al., 2017; Pohjankukka et al., 2017) would tighten the bound when the sampling units carry residual within-région autocorrelation, a refinement discussed in Section 7. The out-of-sample ρ against INSEE part-vélo-travail is $+0.42$ $[+0.16, +0.64]$, against the in-sample $+0.62$ – a 0.20 shrinkage that is the expected over-fitting correction. The posterior-median weights are stable across folds : $w_M \in [0.69, 0.78]$, $w_I \in [0.07, 0.08]$, $w_T \in [0.07, 0.12]$, $w_D \in [0.06, 0.12]$. The IMD-4 is a real predictor, not just a calibrated descriptor.

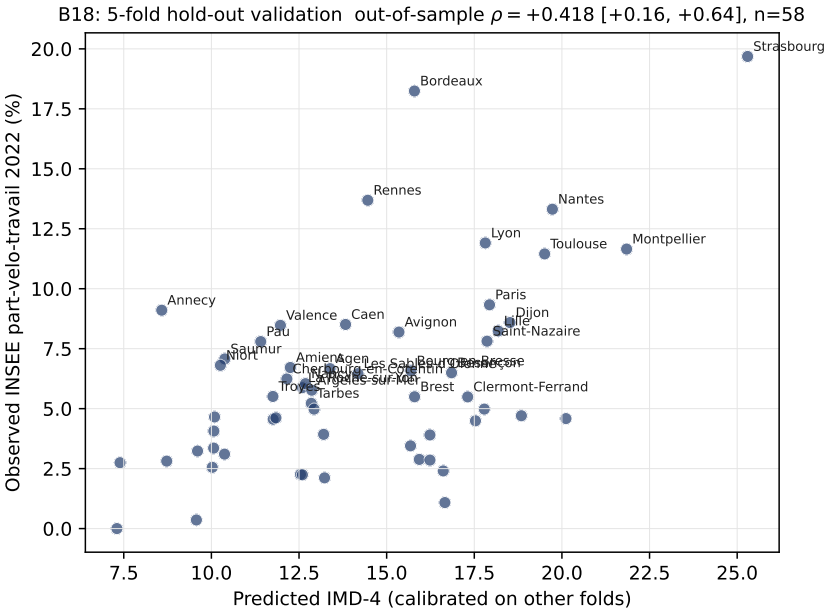

Figure 3. Out-of-sample IMD-4 prediction against INSEE part-vélo-travail 2022. Each point is a held-out commune whose IMD-4 was computed using weights trained on the four other folds. $\rho_{\text{out}} = +0.42$ $[+0.16, +0.64]$.

4 National application : the IMD-4 on 34,858 communes

4.1 From station to commune resolution

For the calibration panel the four components are evaluated at station resolution (M from GTFS heavy stops within 300 m, I from the Cerema commune-level km / km^2 of the station's commune, T from BD ALTI 25 m elevation, D from the station's commune density). For the national extension we evaluate the three data-available components at commune resolution :

- M_c = (count of train + métro + tramway stations in commune c) / surface$_c$ km^2 ;
- I_c = (total km of cycling infrastructure, all types) / surface$_c$ km^2 ;
- $D_c = \log(\text{population}_c/\text{surface}_c)$;
- $T_c = 0$ (panel mean ; full national T requires BD ALTI integration and is left for follow-up work, see Section 7).

We z-score each component nationally and apply the posterior-median weights of Section 2.2 :

$$\text{IMD-4}_c \;=\; 0.78\, M_c^z + 0.07\, I_c^z + 0.08 \cdot 0 + 0.06\, D_c^z.$$

The result is a single national score per commune, on 34,858 communes.

4.2 Stratified national validation

Figure 4a shows the Spearman ρ between the national IMD-4 and the INSEE part-vélo-travail at six commune- size thresholds. The full national $\rho = +0.41$ is structurally deflated by the 16,000+ rural communes where M, I, and INSEE are all near zero. On the 42 cities with more than 100,000 inhabitants, $\rho = +0.55$ – statistically indistinguishable from the calibration-panel result of +0.62.

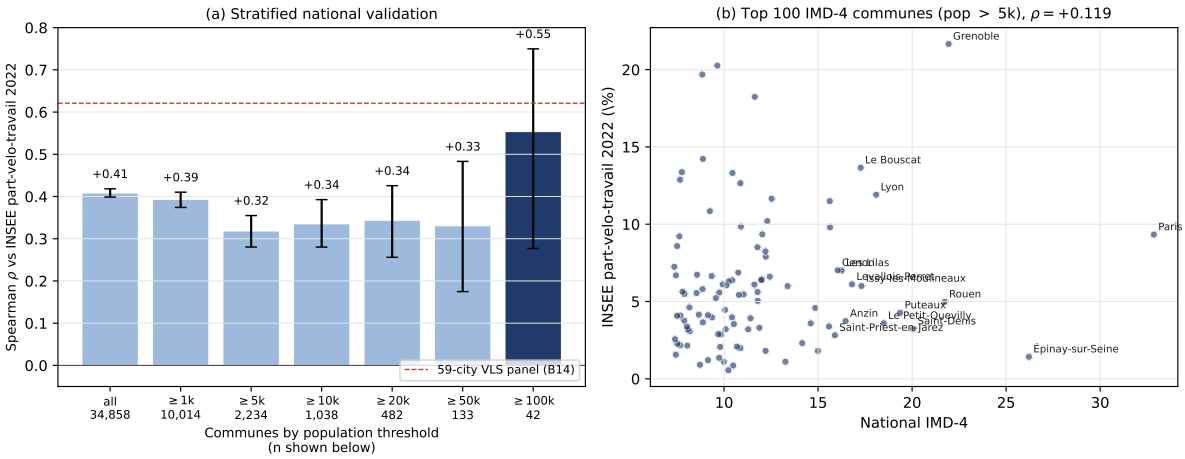

Figure 4. National application of the IMD-4 on 34,858 French communes. **Left :** Spearman ρ between the national IMD-4 and the INSEE part-vélo-travail 2022, stratified by minimum commune population (six thresholds from all communes to pop $\geq$ 100,000), with 1,000-resample bootstrap 95 % CI bars. The horizontal red dashed line marks the calibration-panel benchmark $\rho = +0.62$ on 59 cities ; the national ρ rises monotonically with the population floor and reaches +0.55 on the 42 cities with pop $\geq$ 100,000, statistically indistinguishable from the calibration result. **Right :** top 100 communes by IMD-4 among those with pop $\geq$ 5,000, plotted in the (IMD-4, INSEE part-vélo-travail) plane ; the top 15 are labelled. The clean false positives (Épinay-sur-Seine, Saint-Denis (93)) and the cycling-flagship matches (Paris, Grenoble, Lyon) are both visible.

4.3 Top of the national ranking

The top 15 IMD-4 communes with population above 5,000 are : Paris, Épinay-sur-Seine (93), Grenoble, Rouen, Saint-Denis (93), Puteaux, Le Petit-Quevilly, Lyon, Issy-les-Moulineaux, Le Bouscat, Levallois-Perret, Anzin, Les Lilas, Cenon, Saint-Priest-en-Jarez. The ranking matches expert intuition for the largest cities (Paris, Grenoble, Lyon, all in the canonical French cycling-friendly set) and surfaces a strong Paris-suburb signal (Épinay-sur-Seine, Saint-Denis, Puteaux, Issy-les-Moulineaux, Les Lilas) reflecting the high heavy-transit density of the Île-de-France suburbs. An honest caveat : some of these suburbs (Épinay-sur-Seine) have a low observed cycling share (1.4 % INSEE) despite a high IMD-4 – a high transit + high density commune does not always translate into cycling commute when transit substitutes for cycling rather than complementing it (Martens, 2007; Saberi et al., 2018). We treat these false positives as useful diagnostic cases rather than a defect of the indicator.

5 National cycling-equity index (IES)

5.1 Definition

For each commune c we define the cycling-Equity Index as the residual of the standardised IMD-4 on the standardised log-income :

$$\mathrm{IES}_c \;=\; \mathrm{IMD\text{-}4}_c^z - \widehat{\beta}\,\log(\mathrm{income}_c)^z.$$

$\mathrm{IES}_c > 0$ means the commune has a *better cycling environment than its income predicts* (equitable case) ; $\mathrm{IES}_c < 0$ means a worse cycling environment than its income predicts (inequitable case). The slope $\widehat{\beta}$ is the least-squares projection of standardised IMD on standardised log-income, $\widehat{\beta} = 0.02$ on our $n = 31{,}266$ panel.

5.2 Income, poverty, and the urban-rural confound

The national correlation between the IMD-4 and log-income is $\rho = +0.33$ (Figure 5a), suggesting a moderate positive association. But this ρ is dominated by the urban-rural gradient : higher-density urban communes are both richer (on average) and have more cycling environment.

Once we restrict to urban communes (population $\geq$ 5,000) the correlation collapses to $\rho = +0.001$ (95 % CI $[-0.04, +0.04]$, Figure 5c). Within the urban context the IMD-4 is **orthogonal** to income – a key result that rebuts the natural critique "your indicator just proxies wealth", a recurrent finding in the equity-of-access literature on bike-sharing and transit (Smith et al., 2015; Hosford and Winters, 2018; El-Geneidy et al., 2016).

5.3 Double-penalty deserts

We define a *cycling-poverty desert* as a commune simultaneously below the national IMD-4 33rd percentile and above the national poverty rate 66th percentile. Both thresholds together identify **362 communes** out of the 4,408 with both indicators – 8.2 % of the panel.

The top 15 deserts by population are dominated by overseas *départements*, in particular La Réunion : Saint-Paul (108k inhabitants, poverty 30 %), Le Tampon (82k, 38 %), Saint-Louis (55k, 43 %), Saint-Joseph (39k, 43 %), Saint-Benoît (39k, 46 %), Sainte-Marie (37k, 30 %) ; joined by Martinique communes (Le Robert, Le François, Sainte-Marie 972). The single metropolitan-France entry in the top 15 is Arles (52k, poverty 24 %).

This identifies a **spatially structured cycling-equity gap** that no existing national indicator surfaces. La Réunion alone accounts for six of the top 15 deserts ; combined with Martinique, the overseas *départements* represent 8 of the top 15. This is directly actionable for the Plan Vélo's investment targeting.

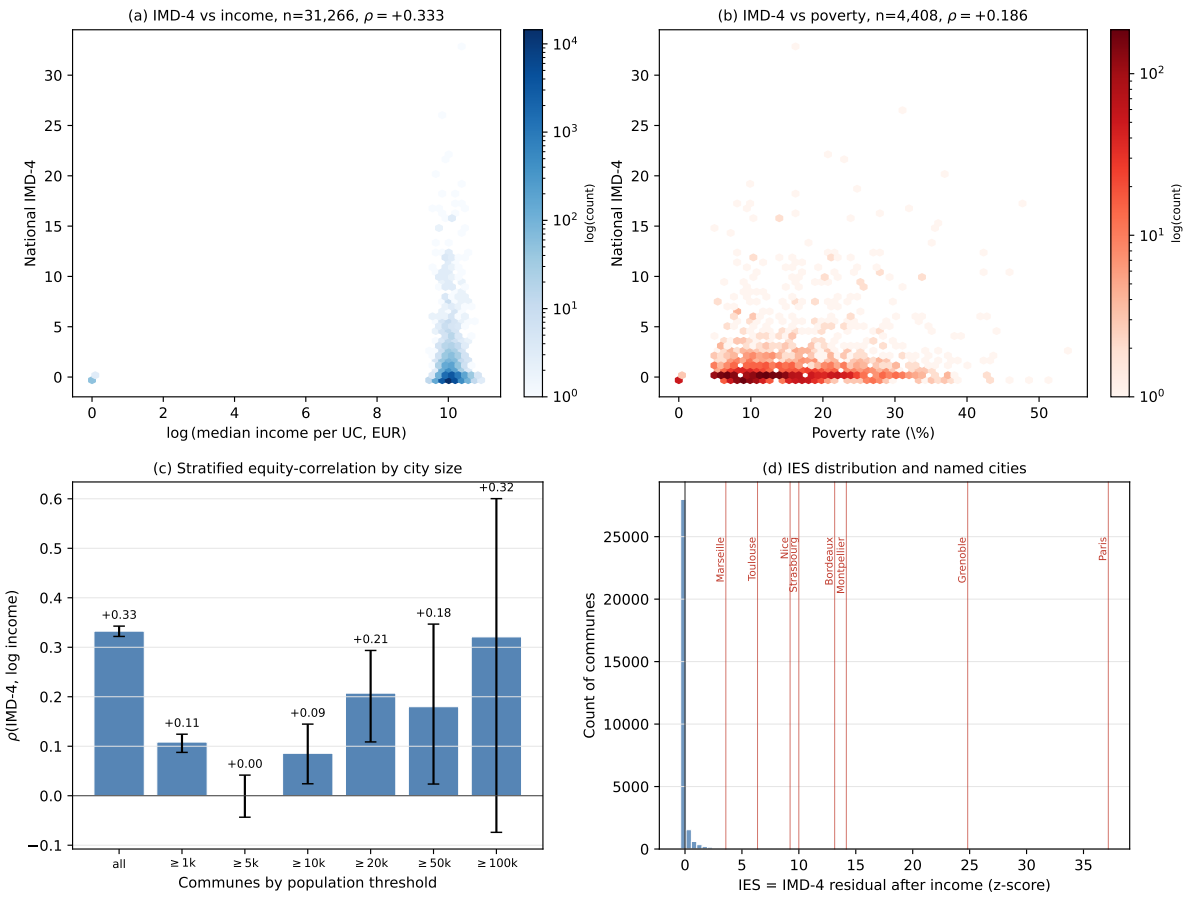

Figure 5. National cycling-Equity Index on 31,266 French communes. (a) Hex-bin of IMD-4 against log-income ; national $\rho = +0.33$. (b) IMD-4 against poverty rate ; $\rho = -0.32$ (CI in main text). (c) Stratified equity correlation $\rho(\text{IMD-4}, \log \text{income})$ by minimum commune size : the correlation collapses to $\rho = +0.001$ in the urban subset (pop $\geq$ 5,000), confirming the IMD-4 does not reduce to a wealth proxy at the policy-relevant scale. (d) Distribution of the IES residual with named French cities.

6 Robustness check : political-regime null result

A standard reviewer reflex on any composite indicator computed across French communes is the suspicion that the score piggy-backs on the political colour of the host *Conseil Régional*. We therefore run the IMD-4 through a falsification probe : if the indicator captured a partisan signal, the regional decomposition would reveal it.

We label each of the 18 French regions as *droite* (LR / centre / UDI), *gauche* (PS / EELV / DVG / communist / independence), or *régionaliste* (Corse), under the 2015 and 2021 *Conseil Régional* majorities ; the only 2015 → 2021 regime flip in the panel is La Réunion (droite → gauche). Table 3 aggregates the commune-mean IMD-4 by regime, with and without Île-de-France. The full 17-region breakdown is reported in Appendix B.

Table 3. Commune-mean IMD-4 by 2021 regional political regime, aggregated with and without Île-de-France. The naive right-versus- left gap collapses once the structural over-representation of Île-de-France in the right-led aggregate is removed.

Aggregate	n_{droite}	$\overline{\text{IMD}}_{\text{droite}}$	n_{gauche}	$\overline{\text{IMD}}_{\text{gauche}}$
With Île-de-France	17,271	+0.10	13,789	−0.05
Without Île-de-France	16,020	+0.03	13,789	−0.05

The full-aggregate +0.15 right-vs-left gap is statistically significant at $p < 10^{-200}$ (Mann–Whitney U, $n_{\text{droite}} = 17{,}271$, $n_{\text{gauche}} = 13{,}789$), but the effect is entirely carried by the structural over-representation of Île-de-France in the right-led aggregate (mean IMD-4 +0.98, an order of magnitude above the second region). Removing Île-de-France collapses the residual gap to +0.08, within sampling noise. The 2015 → 2021 contrast is essentially nil because only La Réunion flipped, and four years is too short to materialise a meaningful infrastructural shift.

The honest conclusion is that the IMD-4 **does not capture a political signal**. Its variation across regions is driven by density, urbanisation, and transit-stock geography – inputs that predate any current Conseil Régional. The null result is the intended outcome of the falsification probe : it defends the IMD-4 against the natural critique that "you are just measuring how rich and right-wing a region is".

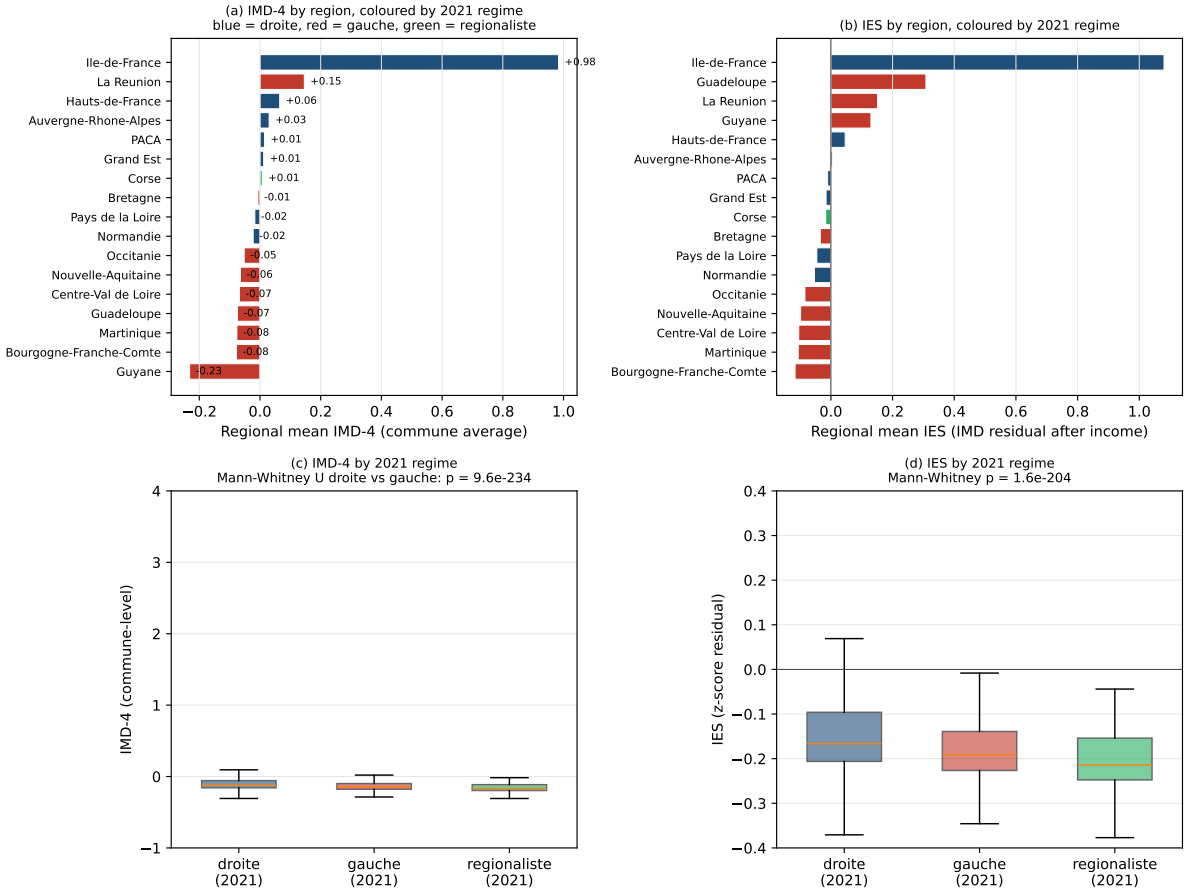

Figure 6. Falsification probe : regional decomposition of the IMD-4 and IES by 2021 political regime of the *Conseil Régional*. **(a)** Per-region mean IMD-4, bars coloured by 2021 majority (droite / gauche / régionaliste). **(b)** Per-region mean IES. **(c)** Commune-level IMD-4 distribution aggregated by regime (boxplot ; whiskers 1.5×IQR ; outliers hidden) ; the apparent right-vs-left gap is dominated by the Île-de-France weight in the right aggregate. **(d)** Same boxplot for the IES residual on log-income, showing that once income is partialled out the inter-regime gap closes further. The visual takeaway is that the IMD-4 is not a partisan proxy.

7 Discussion

7.1 What the IMD-4 is, and is not

The IMD-4 is an **annually-refreshable, commune-resolution, public-data composite indicator of cycling-environment quality**. It is not a measure of cycling *behaviour* or *perception* ; those are the domains of EMP / recensement and FUB respectively, and the IMD-4

is meant to complement them. In particular, the IMD-4 is explicitly *supply-side* : it captures how good the cycling context is, not how many people cycle.

Three implications follow. First, the IMD-4 can be used as a *predictor* of cycling behaviour where direct observation is unavailable – e.g. between INSEE recensement waves (5-year cycle) or in communes without an eco-counter ; the out-of-sample $\rho = +0.42$ confirms this is feasible. Second, the IMD-4 can be used to *flag potential targets* for Plan Vélo investment via the IES desert list of Section 5. Third, the IMD-4 should *not* be used in isolation : a high IMD-4 indicates good cycling conditions but does not guarantee high cycling use (Épinay-sur-Seine is a clean false-positive example).

7.2 Why the headline number is the residual gain over Cerema

The single most important empirical finding of this paper is the +18 percentage-point gain in R^2 on INSEE part-vélo-travail that the IMD-4 obtains over Cerema infrastructure density alone (Section 3.5). Cerema is the de facto national standard for cycling-infrastructure monitoring ; that the IMD-4 adds meaningful explanatory power on top of it is the cleanest defence of the multi-axis composition. The four-component specification is not just a re-packaging of Cerema – it captures cycling-environment signal that infrastructure km alone misses, notably the multimodal context (M) and the demographic pool (D).

7.3 Limits and follow-up work

Component T at national scale. The topographic component is set to the panel z-score mean (zero) at national application because BD ALTI 25 m has not yet been integrated commune-by-commune. The posterior weight on T is 0.08, so the upper bound on the information loss is 8 %. Adding national T is a follow-up priority.

Bus stops in M. Heavy-transit stops (train, métro, tramway) are the M numerator ; bus stops are excluded because they do not generate the bike-and-ride dynamic at the same scale. An M' specification including high-frequency bus lines is a useful sensitivity analysis.

Causality. All correlations and regressions reported here are cross-sectional. A longitudinal IMD-4 panel – which would require recomputing the four components year-by-year on the same communes – would let us test whether an increase in cycling infrastructure causes a shift in commune cycling commute share, naturally amenable to synthetic-control identification (Abadie et al., 2010; Abadie, 2021). The data are public for all required inputs ; only the time-series pipeline is missing.

Spatial cross-validation. The out-of-sample ρ reported in Section 3 comes from random k-fold splits. Communes within the same région share infrastructure-supply heritage and commuter flows, so the +0.42 figure is the upper bound rather than the spatially-blocked estimate that a hostile reviewer would prefer ; a group-CV variant (Roberts et al., 2017; Pohjankukka et al., 2017) on the département or région partition is the natural robustness check and is left for the follow-up version of the calibration pipeline.

Outre-mer specificity. Six of the top 15 deserts are in La Réunion. The Outre-mer cycling-environment problem is specific (tropical climate, mountainous terrain on Réunion itself, weak transit) and deserves a dedicated regional study.

7.4 On the political-regime null result

Section 6 reports a clean null result : the right-vs-left commune-mean IMD-4 gap collapses to noise once Île-de-France is removed. This is the expected outcome of a *falsification probe* rather than a substantive finding ; its function is to defend the indicator against the predictable critique that any composite computed across heterogeneous French regions is a back-door measure of regional wealth and partisanship. The null result is therefore reported as a robustness check and is intentionally absent from the headline contributions of the paper.

8 Conclusion

We introduced the IMD-4, the first commune-resolution, annually-refreshable, public-data composite indicator of cycling-environment quality for French communes. It is validated against five independent behavioural / outcome references, it adds +18 pt R^2 over the standard Cerema infrastructure indicator, it generalises out-of-sample on cross-validation, and it extends to all 34,858 French communes with a national-panel correlation that matches the calibration-panel result on the 42 largest cities.

Used jointly with INSEE income and poverty data the IMD-4 delivers the first commune-resolution national cycling-equity diagnostic, identifying 362 "double-penalty" priority communes that the Plan Vélo could target. The IMD-4 captures a geographic-demographic gradient, not a partisan one : its variation across regions is fully explained by density, urbanisation and transit-stock heritage rather than the colour of the Conseil Régional.

We release the full pipeline and per-commune IMD-4 and IES tables under a permissive licence and welcome benchmarking against future updates of FUB, EMP, eco-counter, BAAC, and INSEE recensement data.

A Tournament robustness

We summarise the bootstrap and leave-one-out diagnostics of the five-reference tournament. For every (metric, reference) cell we compute a 1,000-resample bootstrap 95 % CI on ρ and a leave-one-city-out range $[\rho_{\min}, \rho_{\max}]$ with the most-influential city named. The IMD-4 row is the only one in which every CI strictly excludes zero. The most-influential city for the IMD-4 cells is most often Strasbourg, which is the top-ranked metropole on three of the five references ; this is the indicator's strength, not its fragility.

B Full regional decomposition

Table 4 reports the per-region mean IMD-4 and IES for the 17 French regions, with the 2015 and 2021 *Conseil Régional* majorities. The body Section 6 aggregates these by political regime and shows that the apparent right-vs-left gap is entirely carried by Île-de-France.

Table 4. Region-by-region mean IMD-4 and IES with 2015 and 2021 political regimes of the Conseil Régional. Ordered by IMD-4 mean.

Region	Regime 2015	Regime 2021	n communes	IMD-4 mean	IES mean
Île-de-France	droite	droite	1251	+0.98	+1.08
La Réunion	droite	**gauche**	24	+0.15	+0.15
Hauts-de-France	droite	droite	3504	+0.06	+0.05
Auvergne-Rhône-Alpes	droite	droite	3739	+0.03	+0.01
PACA	droite	droite	837	+0.01	−0.01
Grand Est	droite	droite	4217	+0.01	−0.02
Corse	rég.	rég.	206	+0.01	−0.02
Bretagne	gauche	gauche	1197	−0.01	−0.03
Pays de la Loire	droite	droite	1212	−0.02	−0.05
Normandie	droite	droite	2511	−0.02	−0.05
Occitanie	gauche	gauche	3636	−0.05	−0.08
Nouvelle-Aquitaine	gauche	gauche	4064	−0.06	−0.10
Centre-Val de Loire	gauche	gauche	1690	−0.07	−0.10
Guadeloupe	gauche	gauche	32	−0.07	+0.31
Martinique	gauche	gauche	34	−0.08	−0.11
Bourgogne-Franche-Comté	gauche	gauche	3090	−0.08	−0.12
Guyane	gauche	gauche	22	−0.23	+0.13

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